# Akoizo Original Practice - Form 1 - Answer key

### Division B | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Mohs hardness is…**

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. An ordinal scratch scale

B. A linear mass scale

C. A color scale

D. A toughness scale

**Correct: A**

**2. (2 points) Pyrite belongs to…**

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Sulfides

B. Halides

C. Carbonates

D. Native elements

**Correct: A**

**3. (2 points) Metamorphism usually involves…**

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. Solid-state change

B. Complete melting by definition

C. Only loose sediment

D. No chemical reactions

**Correct: A**

**4. (2 points) Same composition, different structure defines…**

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Sorting

B. Streak

C. Cementation

D. Polymorphism

**Correct: D**

**5. (2 points) Fractional crystallization requires…**

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. Only a color change

B. No crystals

C. Sediment compaction

D. Separation of crystals from melt

**Correct: D**

**6. (2 points) An error log should record…**

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Only total score

B. Only elapsed time

C. Only the answer letter

D. The failed reasoning and correction

**Correct: D**

**7. (2 points) Quartz is a…**

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. Sulfide

B. Halide

C. Framework silicate

D. Double-chain silicate

**Correct: C**

**8. (2 points) Sorting describes…**

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Mineral hardness

B. Fossil age

C. Spread of grain sizes

D. Edge smoothness

**Correct: C**

**9. (4 points) Why can streak fail on quartz?**

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

answer: Quartz can scratch the plate rather than yield a diagnostic powder. (4 points).

**10. (4 points) Why is luster insufficient for gold identification?**

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

answer: Several minerals are metallic and yellowish; hardness, streak, density, and other evidence distinguish them. (4 points).

**11. (4 points) Compare contact and regional metamorphism.**

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

answer: Contact commonly emphasizes local intrusive heating; regional involves broad tectonic burial and deformation. (4 points).

**12. (4 points) What is a pseudomorph?**

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

answer: A replacement mineral preserves the external form of an earlier material. (4 points).

**13. (4 points) What do vesicles record?**

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

answer: Gas bubbles trapped during solidification, without uniquely fixing composition. (4 points).

**14. (4 points) Why use multiple images?**

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

answer: Habit, impurities, lighting, and weathering create variable appearances. (4 points).

**15. (4 points) Explain oxygen sharing.**

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have $SiO_2$ rather than $SiO_4$?

answer: A shared oxygen belongs to adjacent tetrahedra and must not be counted twice in the overall ratio. (4 points).

**16. (4 points) Give two preservation factors.**

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

answer: Rapid burial and durable parts can improve preservation, although other conditions matter. (4 points).

**17. (4 points) Name one alternative-evidence check.**

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

answer: For suspected calcite versus quartz, combine hardness with cleavage or a supplied carbonate reaction observation. (4 points).

**18. (4 points) Explain a mineral use through a property.**

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

answer: Copper’s electrical conductivity supports wiring; hard minerals can serve as abrasives. (4 points).